Linear Flow: Diffusivity Equation Derivation and Solution

Convention: flow in the **−x direction**, mass balanced over a time interval **Δt**. Every step is written out.

Part 0: Setup and assumptions

Geometry: a rectangular block of porous rock with width w, height h, and flow only along x. The cross-sectional area is

$$A = w\,h$$

We study a thin slice between x and x + Δx. Its bulk volume is

$$\Delta V = w\,h\,\Delta x$$

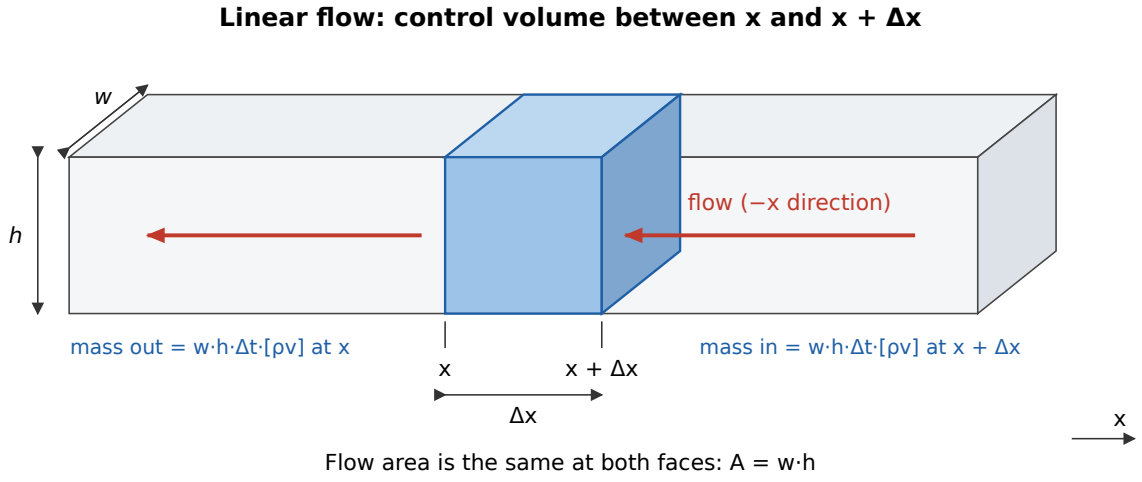


Figure 1. Linear flow. Fluid enters the slice at x + Δx and leaves at x. The flow area w·h is the same on both faces.

Assumptions:

1. Flow in one direction only (x)
2. Single-phase, slightly compressible fluid
3. Homogeneous rock: k constant. Fluid viscosity μ constant
4. Darcy's law applies
5. Horizontal flow (no gravity)
6. Isothermal
7. Small pressure gradients

Variables:

- ρ = fluid density
- v = Darcy velocity (volume flow rate per unit area)
- φ = porosity
- p = pressure
- c = fluid compressibility
- c_f = rock (pore) compressibility

Part 1: Mass of fluid crossing a face

Mass = density × volume. The volume of fluid crossing area A at velocity v during time Δt is A·v·Δt. Multiply by density:

$$\text{mass} = \rho\, A\, v\, \Delta t = \rho\, v\, w\, h\, \Delta t$$

Units check: (kg/m³)·(m/s)·m²·s = kg ✓

Part 2: Mass in, mass out, mass accumulated

Fluid moves in the −x direction. It **enters** the slice through the right face at x + Δx and **leaves** through the left face at x.

Mass in (through the face at x + Δx):

$$(\text{ mass })_{in} = w\, h\, \Delta t\ [\, \rho v\,]_{x+\Delta x}$$

Mass out (through the face at x):

$$(\text{ mass })_{out} = w\, h\, \Delta t\ [\, \rho v\,]_{x}$$

Mass accumulated inside the slice. The mass stored in the slice at any time is:

$$\text{mass stored} = (\text{ pore volume }) \times \rho = (\, \phi\, \Delta V)\ \rho = \Delta V\, (\, \phi \rho\,)$$

The accumulation is the change in stored mass between time t and t + Δt:

$$(\text{ mass })_{acc} = \Delta V [\, (\, \phi \rho\,)_{t+\Delta t} - (\, \phi \rho\,)_{t}]$$

Part 3: Mass balance

Conservation of mass says matter is neither created nor destroyed:

$$(\text{ mass })_{in} - (\text{ mass })_{out} = (\text{ mass })_{acc}$$

Substitute the three expressions:

$$w\,h\,\Delta t\,\left[\,\rho v\,\right]_{x+\Delta x} - w\,h\,\Delta t\,\left[\,\rho v\,\right]_{x} = w\,h\,\Delta x\left[\,\left(\,\phi\rho\,\right)_{t+\Delta t} - \,\left(\,\phi\rho\,\right)_{t}\right]$$

Factor w·h·Δt on the left:

$$w\,h\,\Delta t\left(\,\left[\,\rho v\,\right]_{x+\Delta x} - \,\left[\,\rho v\,\right]_{x}\right) = w\,h\,\Delta x\left[\,\left(\,\phi\rho\,\right)_{t+\Delta t} - \,\left(\,\phi\rho\,\right)_{t}\right]$$

Part 4: Divide and take limits

Divide both sides by w·h·Δx·Δt.

- Left side: w·h cancels, Δt cancels, leaving Δx in the denominator.
- Right side: w·h cancels, Δx cancels, leaving Δt in the denominator.

$$\frac{\left[\,\rho v\,\right]_{x+\Delta x} - \,\left[\,\rho v\,\right]_{x}}{\Delta x} = \frac{\left(\,\phi\rho\,\right)_{t+\Delta t} - \,\left(\,\phi\rho\,\right)_{t}}{\Delta t}$$

This is where Δt and w·h disappear from the equation.

Now let Δx → 0 and Δt → 0. By the definition of the derivative:

$$\lim_{\Delta x \to 0} \frac{f\left(\,x+\Delta x\,\right) - f\left(\,x\,\right)}{\Delta x} = \frac{\partial f}{\partial x}$$

So the left side becomes ∂(ρv)/∂x and the right side becomes ∂(φρ)/∂t. This is the continuity equation, eq. (1):

$$\frac{\partial\left(\,\rho v\,\right)}{\partial x} = \frac{\partial\left(\,\phi\rho\,\right)}{\partial t} \qquad (\,1\,)$$

Part 5: Darcy’s law

Darcy’s law says flow goes from high pressure to low pressure at a rate proportional to the pressure gradient:

$$v = -\frac{k}{\mu}\frac{\partial p}{\partial s}$$

where s is the direction of flow. Here the flow is in the **−x direction**, so s = −x and ∂/∂s = −∂/∂x:

$$v = -\frac{k}{\mu}\left(-\frac{\partial p}{\partial x}\right) = +\frac{k}{\mu}\frac{\partial p}{\partial x} \qquad (\,2\,)$$

Sanity check: if pressure increases with x, then ∂p/∂x > 0 and v > 0. Fluid moves from high pressure (large x) toward low pressure (small x), which is the −x direction

✓

Part 6: Combine continuity and Darcy

Substitute (2) into the left side of (1):

$$\rho v = \rho \frac{k}{\mu} \frac{\partial p}{\partial x}$$

$$\frac{\partial}{\partial x}\left(\rho \frac{k}{\mu} \frac{\partial p}{\partial x} \right) = \frac{\partial \left(\phi \rho \right)}{\partial t}$$

Since k and μ are constants, they can be pulled outside the derivative:

$$\frac{k}{\mu} \frac{\partial}{\partial x}\left(\rho \frac{\partial p}{\partial x} \right) = \frac{\partial \left(\phi \rho \right)}{\partial t} \qquad (3)$$

Now we must deal with ρ and φ, which both depend on pressure. For that we need compressibility.

Part 7: Compressibility definitions

Fluid compressibility: the fractional change in density per unit pressure change:

$$c = \frac{1}{\rho} \frac{\partial \rho}{\partial p}$$

Multiply both sides by ρ:

$$\frac{\partial \rho}{\partial p} = c \rho \qquad (4)$$

Rock compressibility: the fractional change in porosity per unit pressure change:

$$c_f = \frac{1}{\phi} \frac{\partial \phi}{\partial p}$$

$$\frac{\partial \phi}{\partial p} = c_f \phi \qquad (5)$$

Chain rule. Since ρ and φ depend on x and t only through p(x,t):

$$\frac{\partial \rho}{\partial x} = \frac{\partial \rho}{\partial p} \frac{\partial p}{\partial x} = c \rho \frac{\partial p}{\partial x} \qquad (6)$$

$$\frac{\partial \rho}{\partial t} = \frac{\partial \rho}{\partial p} \frac{\partial p}{\partial t} = c \rho \frac{\partial p}{\partial t} \qquad (7)$$

$$\frac{\partial \phi}{\partial t} = \frac{\partial \phi}{\partial p}\frac{\partial p}{\partial t} = c_f \, \phi \, \frac{\partial p}{\partial t} \qquad (8)$$

Part 8: Right-hand side of (3) (accumulation term)

Apply the product rule, ∂(ab)/∂t = a ∂b/∂t + b ∂a/∂t:

$$\frac{\partial \, (\, \phi \rho \,)}{\partial t} = \phi \, \frac{\partial \rho}{\partial t} + \rho \, \frac{\partial \phi}{\partial t}$$

Substitute (7) and (8):

$$= \phi \, (c \, \rho \, \frac{\partial p}{\partial t}) + \rho \, (c_f \, \phi \, \frac{\partial p}{\partial t})$$

Factor out φ ρ ∂p/∂t:

$$= \phi \, \rho \, \left(\, c + c_f \, \right) \, \frac{\partial p}{\partial t}$$

Define **total compressibility** c_t = c + c_f:

$$\frac{\partial \, (\, \phi \rho \,)}{\partial t} = \phi \, \rho \, c_t \, \frac{\partial p}{\partial t} \qquad (9)$$

Part 9: Left-hand side of (3) (flow term)

Apply the product rule to ρ ∂p/∂x, treating ρ as a and ∂p/∂x as b:

$$\frac{\partial}{\partial x}\left(\rho \, \frac{\partial p}{\partial x} \right) = \rho \, \frac{\partial^2 p}{\partial x^2} + \frac{\partial p}{\partial x}\frac{\partial \rho}{\partial x}$$

Substitute (6), ∂ρ/∂x = c ρ ∂p/∂x:

$$= \rho \, \frac{\partial^2 p}{\partial x^2} + \frac{\partial p}{\partial x} \cdot c \, \rho \, \frac{\partial p}{\partial x} = \rho \, \frac{\partial^2 p}{\partial x^2} + c \, \rho \left(\frac{\partial p}{\partial x} \right)^2$$

Factor out ρ:

$$= \rho \left[\frac{\partial^2 p}{\partial x^2} + c \left(\frac{\partial p}{\partial x} \right)^2 \right] \qquad (10)$$

Part 10: Put both sides together

Substitute (10) and (9) into (3):

$$\frac{k}{\mu}\rho\left[\frac{\partial^2 p}{\partial x^2}+c\left(\frac{\partial p}{\partial x}\right)^2\right]=\phi\,\rho\,c_t\,\frac{\partial p}{\partial t}$$

Divide both sides by ρ (it cancels):

$$\frac{k}{\mu}\left[\frac{\partial^2 p}{\partial x^2}+c\left(\frac{\partial p}{\partial x}\right)^2\right]=\phi\,c_t\,\frac{\partial p}{\partial t}$$

Multiply both sides by μ/k:

$$\frac{\partial^2 p}{\partial x^2}+c\left(\frac{\partial p}{\partial x}\right)^2=\frac{\phi\,\mu\,c_t}{k}\,\frac{\partial p}{\partial t}\qquad(11)$$

Equation (11) is **nonlinear** because of the squared gradient term, so it is hard to solve analytically.

Part 11: Linearization

We compare the two terms on the left of (11):

- Term A: $\partial^2 p/\partial x^2$
- Term B: $c\,(\partial p/\partial x)^2$

For a slightly compressible liquid (oil, water), c is very small, typically 10^{-6} to 10^{-5} psi^{-1}. The pressure gradient is also small away from the wellbore. Term B contains c times a squared gradient, so it is far smaller than Term A.

Example: with $c = 10^{-5}$ psi^{-1} and $\partial p/\partial x = 10$ psi/ft, Term B is $10^{-5} \times 100 = 10^{-3}$ psi/ft^2. Term A is usually much larger.

So we drop Term B, which gives the diffusivity equation:

$$\frac{\partial^2 p}{\partial x^2}=\frac{\phi\,\mu\,c_t}{k}\,\frac{\partial p}{\partial t}\qquad(12)$$

Part 12: Hydraulic diffusivity form

Define

$$\eta=\frac{k}{\phi\,\mu\,c_t}$$

Multiply both sides of (12) by η:

$$\frac{\partial p}{\partial t} = \eta \frac{\partial^2 p}{\partial x^2}$$

SI units check:

$$\eta = \frac{\mathrm{m}^2}{(\,\mathrm{Pa}\cdot\mathrm{s}\,)\,\left(\mathrm{Pa}^{-1}\right)} = \frac{\mathrm{m}^2}{\mathrm{s}}$$

This matches the units of a diffusion coefficient ✓

Oilfield units: with k in md, μ in cp, c_t in psi⁻¹, t in hours:

$$\eta = 0.0002637\,\frac{k}{\phi\,\mu\,c_t} \quad \left[\frac{\mathrm{ft}^2}{\mathrm{hr}}\right]$$

Summary of the logic

Step	What we did	Result
1	Mass in − mass out = mass accumulated	Balance over Δt
2	Divide by w·h·Δx·Δt, take limits	Continuity equation (1)
3	Darcy's law for flow in −x	v = (k/μ) ∂p/∂x
4	Substitute Darcy into continuity	Equation (3)
5	Use c and c_f to rewrite ∂(φρ)/∂t	φ ρ c_t ∂p/∂t
6	Expand ∂(ρ ∂p/∂x)/∂x	Gives the nonlinear c (∂p/∂x)² term
7	Neglect that term (small c, small gradient)	Linear diffusivity equation

Solution for Linear Flow (Constant Rate, Infinite-Acting)

The diffusivity equation becomes useful once it is solved for a real boundary condition. The simplest case is a semi-infinite linear reservoir producing at constant rate through a face at x = 0.

Linear Solution Step 1: Problem statement

Reservoir occupies x ≥ 0, with cross-sectional area A. Fluid flows toward x = 0 (the −x direction), where the rate is the constant q. The equation is

$$\frac{\partial p}{\partial t} = \eta \frac{\partial^2 p}{\partial x^2}$$

Initial condition (uniform pressure) and outer boundary (infinite reservoir):

$$p\left(x, 0\right) = p_i, \qquad p\left(x \to \infty, t\right) = p_i$$

Inner boundary. Darcy's law at the face, with flow in the −x direction, is q = (kA/μ) ∂p/∂x:

$$\frac{\partial p}{\partial x}\Big|_{x=0} = \frac{q\,\mu}{kA}$$

Linear Solution Step 2: Work with the pressure gradient

Differentiate the diffusivity equation with respect to x. Because η is constant, the gradient w = ∂p/∂x satisfies the same equation:

$$\frac{\partial w}{\partial t} = \eta \frac{\partial^2 w}{\partial x^2}$$

The conditions on w are

$$w\left(x, 0\right) = 0, \qquad w\left(x \to \infty, t\right) = 0, \qquad w\left(0, t\right) = W \equiv \frac{q\,\mu}{kA}$$

The initial and outer conditions follow because the pressure is uniform far away and at t = 0.

Linear Solution Step 3: Boltzmann transformation

Combine x and t into one variable and look for w = f(s):

$$s = \frac{x}{2\sqrt{\eta t}}$$

The derivatives of s are

$$\frac{\partial s}{\partial x} = \frac{1}{2\sqrt{\eta t}}, \qquad \frac{\partial s}{\partial t} = -\frac{s}{2t}$$

By the chain rule:

$$\frac{\partial w}{\partial t} = -\frac{s}{2t} f', \qquad \frac{\partial^2 w}{\partial x^2} = \frac{f''}{4\eta t}$$

Substitute into the equation. The factor 1/t cancels and the PDE becomes an ordinary differential equation in s:

$$-\frac{s}{2t}f' = \eta\frac{f''}{4\eta t} \quad \Rightarrow \quad f'' + 2s\,f' = 0$$

Linear Solution Step 4: Solve the ODE

Let g = f'. Then g'/g = −2s, so

$$\ln g = -s^2 + C \quad \Rightarrow \quad f'(s) = C_1 e^{-s^2}$$

The condition w → 0 as x → ∞ (that is, s → ∞) lets us write f(s) as an integral from s to ∞:

$$f(s) = -C_1\int_s^\infty e^{-\sigma^2}\,d\sigma = -C_1\frac{\sqrt{\pi}}{2}\operatorname{erfc}(s)$$

using the definition erfc(s) = (2/√π) ∫ₛ^∞ e^(−σ²) dσ. The condition w(0,t) = W means f(0) = W. Since erfc(0) = 1, this fixes C₁ = −2W/√π and

$$w = \frac{\partial p}{\partial x} = W\operatorname{erfc}\left(\frac{x}{2\sqrt{\eta t}}\right)$$

Linear Solution Step 5: Integrate to get pressure

Since p → p_i at large x and ∂p/∂x = w:

$$p_i - p(x,t) = \int_x^\infty w\,dx' = W\cdot 2\sqrt{\eta t}\int_s^\infty \operatorname{erfc}(\sigma)\,d\sigma$$

Integrate by parts, using d(erfc σ)/dσ = −(2/√π) e^(−σ²):

$$\int_s^\infty \operatorname{erfc}(\sigma)\,d\sigma = \left[\sigma\operatorname{erfc}(\sigma)\right]_s^\infty + \frac{2}{\sqrt{\pi}}\int_s^\infty \sigma\,e^{-\sigma^2}\,d\sigma = -s\operatorname{erfc}(s) + \frac{e^{-s^2}}{\sqrt{\pi}}$$

Putting this together:

$$p_i - p(x,t) = \frac{q\mu}{kA}\left[2\sqrt{\frac{\eta t}{\pi}}\,e^{-x^2/(4\eta t)} - x\operatorname{erfc}\left(\frac{x}{2\sqrt{\eta t}}\right)\right]$$

Check at x = 0. The derivative of the bracket is −erfc(0) = −1 at x = 0, so ∂p/∂x = qμ/(kA), which is the boundary condition ✓.

Linear Solution Step 6: Pressure at the face and what it tells you

At x = 0 the second term vanishes and the exponential equals 1:

$$p_i - p_{face}\,(\,t\,) \;= \frac{2\,q\,\mu}{kA}\sqrt{\frac{\eta\,t}{\pi}}$$

The pressure drop grows as **√t**. This is the signature of linear flow: a plot of p against √t is a straight line. Its slope m gives permeability. With η = k/(φμc_t) the slope is

$$m = \frac{2\,q\,\mu}{A\sqrt{\pi\,k\,\phi\,\mu\,c_t}} \quad \Rightarrow \quad k = \frac{4\,q^2\,\mu}{\pi\,\phi\,c_t\,A^2\,m^2}$$

(SI units.) This is why a half-slope (1/2) line on a log-log pressure-derivative plot indicates linear flow, for example into a fracture or along a channel.

The pressure disturbance reaches a distance of order 2√(ηt). This √(ηt) is the diffusion length scale that governs how far pressure travels in any geometry.

Notes: Common Questions and Nuances

N1. Why c_f for rock and c for fluid?

It is a notation convention. The **f stands for formation** (the rock), not fluid. Fluid compressibility carries a phase subscript in general: c_o (oil), c_w (water), c_g (gas). In the derivation there is a single fluid, so its compressibility is written simply as c. For multiphase flow the total compressibility is

$$c_t = c_f + S_o c_o + S_w c_w + S_g c_g$$

For one liquid phase this reduces to c_t = c + c_f, which is what was used.

N2. Why is porosity multiplied in the accumulation term?

Fluid is stored only in the pore space. The slice has bulk volume ΔV, but only the fraction φ of it is open pore space:

$$\text{pore volume} = \phi\,\Delta V, \qquad \text{mass stored} = \rho\;(\,\phi\,\Delta V\,)\; = \Delta V\,(\,\phi\rho\,)$$

Leaving φ out would treat the whole slice as filled with fluid, as if there were no rock.

Two related points:

- **φ is not in the mass in/out terms.** Darcy velocity v is flow rate per unit *total* (bulk) cross-sectional area, so porosity is already built into it.
- **φ sits inside the time derivative, ∂(φρ)/∂t.** Both ρ and φ change with pressure. Fluid expansion changes ρ (compressibility c). Pore volume change

alters φ (compressibility c_f). The product rule on φρ is what produces c + c_f = c_t. Treating φ as constant would lose the rock compressibility.

2D and 3D Cartesian Diffusivity Equation

The idea is the same as the 1D case, but fluid now crosses **three pairs of faces**. Here velocities are taken positive in the +x, +y, +z directions, so mass enters at the lower face of each pair and leaves at the upper one. The final result does not depend on this choice.

3D Step 1: Control volume and faces

A block with sides Δx, Δy, Δz has volume ΔV = Δx Δy Δz. Faces normal to x have area Δy Δz, faces normal to y have area Δx Δz, faces normal to z have area Δx Δy.

3D Step 2: Mass balance over Δt

Mass crossing each face is ρ A v Δt. Writing in minus out for each direction, and the accumulation Δx Δy Δz [$(\varphi\rho)_{t+\Delta t} - (\varphi\rho)_t$]:

$$\begin{aligned}
&\Delta y \Delta z \Delta t([\rho v_x]_x - [\rho v_x]_{x+\Delta x}) \\
&+\Delta x \Delta z \Delta t([\rho v_y]_y - [\rho v_y]_{y+\Delta y}) \\
&+\Delta x \Delta y \Delta t([\rho v_z]_z - [\rho v_z]_{z+\Delta z}) \\
&= \Delta x \Delta y \Delta z[(\phi\rho)_{t+\Delta t} - (\phi\rho)_t]
\end{aligned}$$

3D Step 3: Divide and take limits

Divide by Δx Δy Δz Δt. In each term on the left, two lengths cancel and one stays in the denominator. Taking all increments to zero:

$$-\frac{\partial(\rho v_x)}{\partial x} - \frac{\partial(\rho v_y)}{\partial y} - \frac{\partial(\rho v_z)}{\partial z} = \frac{\partial(\phi\rho)}{\partial t} \qquad (3D\text{-}1)$$

In vector form this is $-\nabla\cdot(\rho\mathbf{v}) = \partial(\varphi\rho)/\partial t$.

3D Step 4: Darcy's law in each direction

$$v_x = -\frac{k}{\mu}\frac{\partial p}{\partial x}, \qquad v_y = -\frac{k}{\mu}\frac{\partial p}{\partial y}, \qquad v_z = -\frac{k}{\mu}\frac{\partial p}{\partial z}$$

Substituting into (3D-1), the two minus signs cancel and k, μ come out of the derivatives:

$$\frac{k}{\mu}\left[\frac{\partial}{\partial x}\left(\rho\frac{\partial p}{\partial x}\right)+\frac{\partial}{\partial y}\left(\rho\frac{\partial p}{\partial y}\right)+\frac{\partial}{\partial z}\left(\rho\frac{\partial p}{\partial z}\right)\right]=\frac{\partial\left(\,\phi\rho\,\right)}{\partial t}$$

3D Step 5: Expand and linearize

Each term expands as in 1D (product rule, then ∂ρ/∂x = cρ ∂p/∂x):

$$\frac{\partial}{\partial x}\left(\rho\frac{\partial p}{\partial x}\right)=\rho\frac{\partial^2 p}{\partial x^2}+c\rho\left(\frac{\partial p}{\partial x}\right)^2$$

and similarly for y and z. The accumulation term is unchanged, ∂(φρ)/∂t = φρ c_t ∂p/∂t. Cancelling ρ and multiplying by μ/k:

$$\frac{\partial^2 p}{\partial x^2}+\frac{\partial^2 p}{\partial y^2}+\frac{\partial^2 p}{\partial z^2}+c\left[\left(\frac{\partial p}{\partial x}\right)^2+\left(\frac{\partial p}{\partial y}\right)^2+\left(\frac{\partial p}{\partial z}\right)^2\right]=\frac{\phi\mu c_t}{k}\frac{\partial p}{\partial t}$$

Dropping the small c|∇p|² term gives the 3D diffusivity equation:

$$\frac{\partial^2 p}{\partial x^2}+\frac{\partial^2 p}{\partial y^2}+\frac{\partial^2 p}{\partial z^2}=\frac{1}{\eta}\frac{\partial p}{\partial t},\qquad \eta=\frac{k}{\phi\mu c_t}$$

2D case

For a thin reservoir of thickness h with flow only in the x-y plane, the block is Δx × Δy × h. The thickness h appears in every term and cancels, just as w·h did in 1D. The result is

$$\frac{\partial^2 p}{\partial x^2}+\frac{\partial^2 p}{\partial y^2}=\frac{1}{\eta}\frac{\partial p}{\partial t}$$

Two practical notes

Anisotropy. If permeability differs by direction, Darcy's law uses the matching permeability in each direction:

$$k_x\frac{\partial^2 p}{\partial x^2}+k_y\frac{\partial^2 p}{\partial y^2}+k_z\frac{\partial^2 p}{\partial z^2}=\phi\mu c_t\frac{\partial p}{\partial t}$$

Vertical permeability k_z is usually the smallest.

Gravity. In the vertical direction Darcy's law has an extra term, v_z = −(k/μ)(∂p/∂z + ρg) with z measured upward. It was neglected above. For a slightly compressible liquid, ρ varies very little, so the gravity term is nearly constant in space and drops out of the derivative at this level of approximation. Pressure can then be read as pressure referred to a datum. For accurate vertical-flow problems (gas caps, coning), flow potential is used instead of pressure.